Possible and Necessary Popularity in House Allocation PseudoCode

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1 Hypothesis

- Agents give their partial strict linear order. They indicate positions of each object in their ranking.
- There are as many objects as agents.
- We consider only strict linear order.

2 Properties

Proposition 1 (Verify Possibly popular matching)**.** *If* $\exists i, j \in N$ *such that* $\sigma(j) P_i \sigma(i)$ *and* $\exists o \in O$ *such that* $o \succ_j \sigma(j)$ *then return False (the matching is not popular).*

Proposition 2 (Find Possibly popular completion)**.** *If* $\exists i, j \in N$ *such that* $\sigma(j) P_i \sigma(i)$ *for* $o \in O$ *such that* $(\sigma(j), o) \notin P$ *add* $(\sigma(j), o) in \succ_j$

Necessarily popular matching

Proposition 3 (Find Necessarily popular matching)**.** *If* $\exists i, j \in N$ *such that* $\sigma(j) P_i \sigma(i)$ *and* $o \in O$

- $o P_j \sigma(j)$ *or*
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

Proposition 4 (Find Necessarily popular completion)**.** *If* $\exists i, j \in N$ *such that* $(\sigma(i), \sigma(j)), (\sigma(j), \sigma(i)) \notin P_i$ *and* $o \in O$ *such that*

- $o P_j \sigma(j)$
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

Proposition 5. *If at least half of agents have the same fpost, it is not possible to find a popular matching.*

3 Algorithms

Algorithm 1 IsPreferredTo

Input: $o \in O$, $o' \in O$, $P_k \in P$
Output: boolean indicating either o is preferred to o' by agent k

for condition **do**
    operation
end for
return result

Algorithm 2 VerifPossiblePopularMatching

Input: O the set of n objects, N the set of n agents, $P = (P_1, ..., P_n)$ the partial profile such that $P_i = [?, \sigma(i), o, ...]$ is the list of partial preferrence of agent i (? indicates unknown preference and $\sigma(i)$ is the assigned object to i in the matching), $\mu = \{1\colon \sigma(1), ..., n\colon \sigma(n)\}$ is the matching.
Output: boolean indicating either the matching μ is possibly popular, P a completion of the partial profile making μ possibly popular.

for each agent $i \in N$ **do**
    for each preference $p \in P_i$ **do**
        if p is unknown (equal to ?) **then**
            // Is there a completion such as p is the fpost matched to another agent in μ?
            Find a matched fpost (should not be a fpost for more than half agents) and assigned it to p
        else if p is $\sigma(i)$ **then**
            Complete the preference list P_i with the remaining objects.
            Break
        else
            // p is an object with a better rank than $\sigma(i)$
            if p is a fpost matched to another agent **then**
                Continue
            else
                // p is not a matched fpost, the matching μ cannot be popular
                return False, P
            end if
        end if
    end for
end for
return result

4 Examples

Let H an instance of the House Allocation problem such as:

- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i.

- $\mu = \{(Alice, b), (Bob, c), (Amelie, a)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i		**Completions**	
Alice	? a ?	$\boxed{\text{b}}$ a c	$\boxed{\text{b}}$ a c	c a $\boxed{\text{b}}$
Bob	b ? ?	b $\boxed{\text{c}}$ a	b $\boxed{\text{c}}$ a	b $\boxed{\text{c}}$ a
Amelie	? a ?	c $\boxed{\text{a}}$ b	b $\boxed{\text{a}}$ a	b $\boxed{\text{a}}$ c
		not popular	no solution	not popular